Newton's Laws of Motion

Physics 101

Newton's Second Law

The acceleration of an object depends on the net force acting on it and the mass of the object.

A larger net force produces a larger acceleration. For the same force, a larger mass produces a smaller acceleration.

$$F = m\,a$$

Force is measured in newtons, mass in kilograms, and acceleration in metres per second squared.

Worked Example

A 2 kg block is pushed by a net force of 10 N.

Step 1: Write the equation.

$$a = F / m$$

Step 2: Substitute the values.

$$a = 10\ N / 2\ kg = 5\ m/s^2$$

The block accelerates at 5 metres per second squared.